

Solutions for AJC H2 Prelim 2009 Paper 2

Qn.	Solutions	
1	$(1+x^2)\frac{dy}{dx} + xy = \sqrt{1+x^2}$ $(1+x^2)\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} + x\frac{dy}{dx} + y = \frac{1}{2} \times \frac{2x}{\sqrt{1+x^2}}$ $(1+x^2)\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = \frac{x}{\sqrt{1+x^2}}$ When $x=0$, $y=1$, (1) $\Rightarrow \frac{dy}{dx} = 1$, (2) $\Rightarrow \frac{d^2y}{dx^2} = -1$, By Maclaurin's series, $y = f(x) = f(0) + f'(0)x + \frac{f''(0)}{(2!)}x^2 + \dots$ $y = 1 + x + \frac{-1}{(2!)}x^2 + \dots = 1 + x - \frac{1}{2}x^2 + \dots$ Method 1: $\int_0^{0.01} \frac{dy}{dx} dx = [y]_0^{0.01}$ $= \left[1 + x - \frac{1}{2}x^2 + \dots \right]_0^{0.01} = 1 + 0.01 - \frac{1}{2}(0.01)^2 - 1 \approx 0.00995(3s.f) \approx 0.010(3d.p)$ Method 2: $\int_0^{0.01} \frac{dy}{dx} dx = \int_0^{0.01} 1 - x dx = \left[x - \frac{x^2}{2} \right]_0^{0.01} \approx 0.00995(3s.f) \approx 0.010(3d.p.)$ NOTE – Integration done by GC - accepted	
2(i)	$\frac{EE'}{3} = \tan \theta \Rightarrow EE' = \frac{4}{3} \times 3 = 4 \quad \therefore \overrightarrow{E'E} = 4\mathbf{k}$ $\overrightarrow{OF} = \overrightarrow{OE'} + \overrightarrow{E'E} + \overrightarrow{EF} = 3\mathbf{j} + 4\mathbf{k} + 10\mathbf{i} \text{ (Shown)}$	

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2(ii)	$\overrightarrow{DB} = \begin{pmatrix} 10 \\ -6 \\ 0 \end{pmatrix} \quad \overrightarrow{DE} = \begin{pmatrix} 0 \\ -3 \\ 4 \end{pmatrix}$ $\vec{n} = \begin{pmatrix} 10 \\ -6 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -3 \\ 4 \end{pmatrix} = \begin{pmatrix} -24 \\ -40 \\ -30 \end{pmatrix} = -2 \begin{pmatrix} 12 \\ 20 \\ 15 \end{pmatrix}$ $\sin \theta = \frac{\left \begin{pmatrix} 12 \\ 20 \\ 15 \end{pmatrix} \cdot \begin{pmatrix} 10 \\ 3 \\ 4 \end{pmatrix} \right }{\sqrt{769}\sqrt{125}}$ $\theta = 50.7228..^\circ = 50.7^\circ$	
3(a)	$(x+y)\frac{dy}{dx} = x^2 + xy + x + 1 \quad \text{-----(1)}$ Given $z = x + y \Rightarrow \frac{dz}{dx} = 1 + \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{dz}{dx} - 1 \quad \text{-----(2)}$ Sub (2) into (1) : $(z)\left(\frac{dz}{dx} - 1\right) = x^2 + x(z - x) + x + 1$ $(z)\frac{dz}{dx} = z + xz + x + 1$ $(z)\frac{dz}{dx} = z(x+1) + (x+1)$ $(z)\frac{dz}{dx} = (x+1)(z+1) \quad (\text{shown})$ $\int \frac{z}{z+1} dz = \int (x+1) dx$ $\int 1 - \frac{1}{z+1} dz = \frac{(x+1)^2}{2} + c$ $z - \ln z+1 = \frac{1}{2}(x+1)^2 + c$ $x + y - \ln x+y+1 = \frac{1}{2}(x+1)^2 + c$	
3(b)(i)	Speed = $\frac{dx}{dt} = k(8-x)$, k = positive constant At $t = 0, x = 0$, $\frac{dx}{dt} = 10$ $\Rightarrow 10 = 8k \Rightarrow k = \frac{5}{4}$	

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	$\Rightarrow \frac{dx}{dt} = \frac{5}{4}(8-x)$ (shown)	
3(b)(ii)	Integrating wrt t, $\int \frac{1}{8-x} dx = \int \frac{5}{4}$ $-\ln(8-x) = \frac{5}{4}t + C$ $x = 8 - Ae^{-\frac{5}{4}t}$ At $t = 0, x = 0, A = 8 \Rightarrow x = 8 - 8e^{-\frac{5}{4}t}$ At $x = 6, e^{-\frac{5}{4}t} = \frac{1}{4} \Rightarrow t = 1.11\text{hrs}$	
3(b)(iii)	$x = 8 - 8e^{-\frac{5}{4}t}$ When $x = 8$ (i.e. he completes his 8 km jog), $t \rightarrow \infty$ Model predicts infinite amount of time to complete jog, therefore model is NOT suitable.	
4(i)	The sequence converges to α. Thus as $n \rightarrow \infty, x_n \rightarrow \alpha$ and $x_{n+1} \rightarrow \alpha$ $\alpha = \frac{k+2\alpha}{2+\alpha}$ $\Rightarrow 2\alpha + \alpha^2 = k + 2\alpha \Rightarrow \alpha^2 = k \Rightarrow \alpha = +\sqrt{k}, \text{ as } \alpha > 0$	
4(ii)	$x_{n+1} - x_n = \frac{k+2x_n}{2+x_n} - x_n$ $= \frac{k+2x_n-2x_n-x_n^2}{2+x_n} = \frac{k-x_n^2}{2+x_n} = \frac{(\sqrt{k}-x_n)(\sqrt{k}+x_n)}{2+x_n}$ $x_n < \sqrt{k} \Rightarrow \sqrt{k}-x_n > 0$ $\Rightarrow x_{n+1} - x_n > 0 \Rightarrow x_{n+1} > x_n$	
4(iii)	$x_{n+1} - \sqrt{k} = \frac{k+2x_n}{2+x_n} - \sqrt{k}$ $= \frac{k+2x_n-2\sqrt{k}-x_n\sqrt{k}}{2+x_n}$ $= \frac{x_n(2-\sqrt{k})-\sqrt{k}(2-\sqrt{k})}{2+x_n}$ $\Rightarrow x_{n+1} - \sqrt{k} = \frac{(x_n-\sqrt{k})(2-\sqrt{k})}{2+x_n} \text{ (proved)}$ For $x_{n+1} < \sqrt{k}$ whenever $x_n < \sqrt{k}$ for all $n \in \mathbb{Z}^+$. $\Rightarrow x_{n+1} - \sqrt{k} < 0 \text{ whenever } x_n - \sqrt{k} < 0 \text{ for all } n \in \mathbb{Z}^+.$ $\Rightarrow 2 - \sqrt{k} > 0 \Rightarrow 0 < k < 4.$	
4(iv)	If $k=1$ and $x_1=0.9$,	

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	by using GC, $x_2 = 0.966$, $x_3 = 0.988$	
4(v)	From (ii) and (iii), as $x_1 = 0.9 < \sqrt{k} = 1$, we have $x_n < x_{n+1} < \sqrt{k} = 1$ for all $n \in \mathbb{Z}^+$. Thus $x_1 < x_2 < x_3 < \dots < \sqrt{k} = 1$ The sequence $x_1, x_2, x_3, \dots$ is an increasing sequence that converges to 1.	
5(a)	$iz + 2w = 1 \Rightarrow -z + 2iw = i \Rightarrow z = 2iw - i \dots (1)$ $4z + (2-i)w^* = -6 \dots (2)$ Sub (1) into (2) $4(2iw - i) + (2-i)w^* = -6$ Let $w = x + yi$ $8i(x + yi) + (2-i)(x - yi) = -6 + 4i$ $8xi - 8y + 2x - 3yi - xi - y = -6 + 4i$ $(-8y + 2x - y) + (8x - x - 3y)i = -6 + 4i$ Comparing : $-9y + 2x = -6 \Rightarrow -3y + x = -2 \dots (3)$ $7x - 3y = 4 \dots (4)$ Solving (3) & (4) $7(3y - 2) - 3y = 4 \Rightarrow 18y = 18$ $\Rightarrow y = 1 \Rightarrow x = 1$ So $w = 1 + i$ $\Rightarrow z = 2i(1 + i) - i = -2 + i$	
5(b)	$z^5 = 1 \Rightarrow z^5 = e^{2k\pi i}$ $\Rightarrow z = e^{\frac{2k\pi i}{5}}, \quad k = 0, 1, 2, 3, 4$ $w^5 = (w - i)^5$ $\left(\frac{w}{w - i}\right)^5 = 1$ $\frac{w}{w - i} = e^{\frac{2k\pi i}{5}}$ $\Rightarrow w = (w - i)e^{\frac{2k\pi i}{5}}$ $\Rightarrow w = \frac{-ie^{\frac{2k\pi i}{5}}}{1 - e^{\frac{2k\pi i}{5}}}$	

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	$= \frac{-ie^{\frac{2k\pi}{5}i}}{e^{\frac{k\pi}{5}i} \left(e^{\frac{-k\pi}{5}i} - e^{\frac{k\pi}{5}i} \right)}$ $\Rightarrow w = \frac{-ie^{\frac{k\pi}{5}i}}{e^{\frac{k\pi}{5}i} - e^{\frac{-k\pi}{5}i}}$ $\Rightarrow w = \frac{-i \left(\cos\left(\frac{k\pi}{5}\right) + i \sin\left(\frac{k\pi}{5}\right) \right)}{-2i \sin\left(\frac{k\pi}{5}\right)}$ $\Rightarrow w = \frac{1}{2} \left[\cot\left(\frac{k\pi}{5}\right) + i \right], \quad k = 1, 2, 3, 4$	
6(i)	$\frac{10!}{3!5!2!} = 2520$	
6(ii)	Arrange the 3 red and 2 yellow: $\frac{5!}{3! \times 2!} = 10$ Case 1: all the 5 blue flags are together Number of ways = 6C_1 Case 2: 4 blue + 1 blue: Number of ways = ${}^6C_1 \times {}^5C_1$ Case 3: 3 blue + 2 blue Number of ways = ${}^6C_1 \times {}^5C_1$ Case 4: 3 blue + 1 blue + 1 blue Number of ways = ${}^6C_1 \times {}^5C_2$ Total number of ways = $10 ({}^6C_1 + ({}^6C_1 \times {}^5C_1) + ({}^6C_1 \times {}^5C_1) + ({}^6C_1 \times {}^5C_2)) = 1260$ No yellow flag selected. Number of ways = $\binom{8}{4} = 70$ One yellow flag selected. Number of ways = $\binom{8}{3} \binom{2}{1} = 112$ Number of ways required = $\binom{8}{4} + \binom{8}{3} \binom{2}{1} = 182$	

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7(a)(i)	$P(\text{at least 1 diamond}) = P(D\bar{D}) + P(DD) = 2\left(\frac{10}{40} \times \frac{30}{39}\right) + \left(\frac{10}{40} \times \frac{9}{39}\right)$ $= \frac{23}{52} \text{ (proved)}$ <u>Alternative:</u> $P(\text{at least 1 diamond}) = 1 - P(\text{both not diamond})$ $= 1 - \left(\frac{30}{40} \times \frac{29}{39}\right) = \frac{23}{52}$	
7(a)(ii)	P(one '7' is drawn at least one diamond card is drawn) $= \frac{P(D \text{ 7, non-D and non-7}) + P(\text{non-D 7, D non-7}) + P(D \text{ 7, D non-7})}{P(\text{at least one diamond is drawn})}$ $= \frac{2\left(\frac{1}{40} \times \frac{27}{39}\right) + 2\left(\frac{3}{40} \times \frac{9}{39}\right) + 2\left(\frac{1}{40} \times \frac{9}{39}\right)}{\frac{23}{52}} = \frac{21}{115} \text{ or } 0.183$	
7(b)	P(obtaining first '7' on n^{th} draw) = $\frac{1}{10} \left(\frac{9}{10}\right)^{n-1}$ Given: $P(\text{obtaining first '7' on or after the } n^{\text{th}} \text{ draw}) \geq 0.4$ $\frac{1}{10} \left(\frac{9}{10}\right)^{n-1} + \frac{1}{10} \left(\frac{9}{10}\right)^n + \frac{1}{10} \left(\frac{9}{10}\right)^{n+1} + \dots \geq 0.4$ $\frac{1}{10} \left(\frac{9}{10}\right)^{n-1} \left[1 + \frac{9}{10} + \left(\frac{9}{10}\right)^2 + \dots\right] \geq 0.4$ $\frac{1}{10} \left(\frac{9}{10}\right)^{n-1} \left(\frac{1}{1 - \frac{9}{10}}\right) \geq 0.4$ $\left(\frac{9}{10}\right)^{n-1} \geq 0.4$	

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	$n - 1 \leq 8.6967 \Rightarrow n \leq 9.6967$ Greatest value of $n = 9$.										
8(i)	There are a total of 400 employees. For sample of 50, $\frac{400}{50} = 8$ That is, after randomly choosing an employee in the first group of 8, every 8 th employee on the list is subsequently chosen. Hence the level 1 group in the marketing department which has 7 employees may not be represented.										
8(ii)	A stratified sample is obtained by randomly choosing $\frac{x}{400} \times 50$ employees (rounded to the nearest integer) for each of the 12 strata above where x is the number of employees in each stratum. <table border="1" data-bbox="451 655 1146 768"> <thead> <tr> <th>Skill level</th><th>Level 1</th><th>Level 2</th></tr> </thead> <tbody> <tr> <td>Male</td><td>45</td><td>126</td></tr> <tr> <td>Female</td><td>75</td><td>154</td></tr> </tbody> </table>	Skill level	Level 1	Level 2	Male	45	126	Female	75	154	
Skill level	Level 1	Level 2									
Male	45	126									
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8(ii)	No of level 1 male employees chosen = $\frac{45}{400} \times 50 = 5.625 \approx 6$ No of level 2 male employees chosen = $\frac{126}{400} \times 50 = 15.75 \approx 16$ No of level 1 female employees chosen = $\frac{75}{400} \times 50 = 9.375 \approx 9$ No of level 2 female employees chosen = $\frac{154}{400} \times 50 = 19.25 \approx 19$ After deciding the number chosen for each stratum, these employees are chosen using random sampling.										
9(i)	Let X = number of complaints in one week Let Y = number of complaints in 3 weeks. $X \sim Po(2)$ and $Y \sim Po(6)$ $X + Y \sim Po(8)$ Required probability = $P(Y = 3 / X + Y = 3)$ $= \frac{P(Y = 3).P(X = 0)}{P(X + Y = 3)}$ $= 0.4218... = 0.422$										
9(ii)	Required probability = $P(X = 0).[P(X = 1)]^3.[P(X > 1)]^2 \times \frac{6!}{3! \times 2!}$ $= 0.0568.$										
9(iii)	Let W = number of complaints in n weeks $W \sim Po(2n)$ Since $n > 5 \Rightarrow 2n > 10$ Therefore $W \sim N(2n, 2n)$ $P(W > 3n) < 0.1 \Rightarrow P(W \geq 3n + 1) < 0.1$										

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	$\Rightarrow P(W \geq 3n + 0.5) < 0.1 \quad (\text{continuity correction})$ $\Rightarrow P\left(Z \geq \frac{3n + 0.5 - 2n}{\sqrt{2n}}\right) < 0.01$ $\Rightarrow P\left(Z \geq \frac{n + 0.5}{\sqrt{2n}}\right) < 0.01$ $\Rightarrow \frac{n + 0.5}{\sqrt{2n}} > 2.3263$ From GC, $n > 9.797$ Least $n = 10$	
10(i)	Let X = number of customers who pay by credit card in sample of 4 Let Y = number of customers who pay by credit card in sample of 15 $X \sim B(4, 0.3), \quad Y \sim B(15, 0.3)$ Required probability $= P(X = 1)P(Y > 4) \times 0.3$ $= P(X = 1)[1 - P(Y \leq 4)] \times 0.3$ $= 0.059827... = 0.0598$	
10(ii)	Let $\bar{Y}$ = average number of customers who pay by credit card $Y \sim B(15, 0.3)$ $E(Y) = 15(0.3) = 4.5 \quad \text{and} \quad \text{Var}(Y) = 15(0.3)(0.7) = 3.15$ By CLT, $\bar{Y} \sim N\left(4.5, \frac{3.15}{50}\right)$ $P(4 < \bar{Y} < a) = 0.95$ $\Rightarrow P(\bar{Y} < a) - P(\bar{Y} < 4) = 0.95$ $\Rightarrow P(\bar{Y} < a) = 0.95 + P(\bar{Y} < 4)$ $\Rightarrow P(\bar{Y} < a) = 0.97318$ $\Rightarrow a = 4.98$	
11(i)	$H_0 : \mu = 60$ $H_1 : \mu > 60$	
11(ii)	$\bar{x} = 63.1, \quad s^2 = \frac{20}{19} \left(\frac{1096}{20} - (3.1)^2 \right) = 47.5684$ $s = 6.897$ Test statistic: $T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t(n-1)$ i.e. $T \sim t(19)$ We reject H_0 if $p < 0.05$ From GC, $p = 0.0294 < 0.05$, we reject H_0 and conclude at 5% sig level that there is sufficient evidence to support the resident's claim.	

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11(iii)	5% level of significance means that: There is a probability of 0.05 of concluding that the resident's claim is correct (justified) when in fact it is not correct (justified). <u>OR</u> There is a probability of 0.05 of concluding that the mean speed of the cars is more than 60km/h when in fact it is not more than 60 km/h.	
11(iv)	Assuming H_0 is true, $Z = \frac{\bar{X} - 60}{\sigma/\sqrt{n}} \sim N(0,1)$ To reject H_0 at 5% level of significance, $\frac{\bar{x} - 60}{24/\sqrt{25}} > 1.645$ $\bar{x} > 67.9$	
11(v)	If the mean speed $\bar{x} = 65 \text{ km/h} < 67.9$, we do not reject H_0 and conclude that there is insufficient evidence to conclude that the average speed of vehicles is greater than 60 km/h .	
12(i)	$\bar{m} = 350, \quad \bar{l} = \frac{1306 + a}{6}$ Substitute them into regression line of l on m, $\frac{1306 + a}{6} = 0.15257(350) + 210.6$ $\Rightarrow a = 277.997$ ie $a = 278$ (shown) From GC, $r = 0.991$	
12(ii)	There is an increase of 0.15257 mm for each gram added When there is no mass attached, the length of the string is 210.6 mm	
12(iii)	The string will break if too much weight is added. <u>OR</u> The string may lose its elasticity if too much weight is added and the nature of the relationship would change.	
12(iv)	$l = 0.15257m + 210.6$ $\Rightarrow l + b = 0.15257(100M) + 210.6$ $\Rightarrow l = 0.15257(100M) + 210.6 - b$ Therefore regression coefficient of L on $M = 100(0.15257) = 15.257$	
12(v)	The product moment correlation remains unchanged when data is scaled or translated. Therefore correlation coefficient of L and M is 0.991	

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13(i)	Let X = duration of calls to City A. Let Y = duration of calls to City B. $X \sim N(8, 2.5^2)$ and $Y \sim N(10, 3.2^2)$ $X_1 + X_2 - 3X_3 \sim N(2(8) - 3(8), 2(2.5)^2 + 3^2(2.5)^2)$ $\Rightarrow X_1 + X_2 - 3X_3 \sim N(-8, 68.75)$ $P(X_1 + X_2 - 3X_3 > 3)$ $= P(X_1 + X_2 - 3X_3 > 3) + P(X_1 + X_2 - 3X_3 < -3)$ $= 0.0923122 + 0.726753$ $= 0.819$	
13(ii)	Total cost $C = 22(X_1 + X_2) + 30Y + 30$ $E(C) = E(22(X_1 + X_2) + 30Y + 30)$ $= E(22(2)(8) + 30(10) + 30) = 682$ Expected amount of money = 682 cents or \$6.82	
13(iii)	$Var(C) = Var(22(X_1 + X_2) + 30Y + 30)$ $= Var(22^2(2)(2.5)^2 + 30^2(3.2)^2) = 15266$ $C \sim N(682, 15266)$ $P(C > 850) = 0.0869606... = 0.0870$ Let W = number of weeks she spends more than \$8.50 $W \sim B(50, 0.086961)$ or $W \sim B(50, 0.0870)$ $np = 4.3481 < 5$ and $n = 50$ is large $\Rightarrow W \sim Po(4.3481)$ or $\Rightarrow W \sim Po(4.35)$ approximately $P(3 < W < 8) = P(4 \leq W \leq 7)$ $= P(W \leq 7) - P(W \leq 3)$ $= 0.92539 - 0.36856$ $= 0.55683... = 0.557$ <u>OR</u> $P(3 < W < 8) = P(4 \leq W \leq 7)$ $= P(W \leq 7) - P(W \leq 3)$ $= 0.92552 - 0.36823$ $= 0.55701... = 0.557$	